

1. Given the relation $\{(2,3),(4,k),(4,2),(5,1)\}$. The **product** of all the distinct numbers in the **range** of the relation shown is 60. Find the value of k .
2. Let $i = \sqrt{-1}$. Let k and w represent positive integers with $k < w$. If $(3i)(ki)(wi) = -12i$, find the value of k .
3. The vector $(5.1, 3.4)$ is perpendicular to the vector $(-6.55, k)$. Find the value of k . Express your answer as an **exact decimal**.
4. The sum of the infinite geometric series of real terms whose first term is k is $\frac{2k}{k+1}$. If $k > 0$ and the second term is k^2 , find the value of k . Express your answer as a common fraction reduced to lowest terms.
5. Find the length of the minor axis of the conic section whose equation is $5x^2 - 10x + 9y^2 - 72y + 104 = 0$.
6. Let $i = \sqrt{-1}$ and let k and w represent real numbers. If $(k + wi)(9 - 2i) = x + 7i$ is solved for x , then $x = 74 - 14i$. Find the value of $(k + w)$.
7. Bob, Christie, and Judy repeatedly take turns tossing a fair, standard cubical die. Bob begins, Christie always follows Bob, and Judy always follows Christie. Find the probability that Christie will be the first to toss a die that lands so that its uppermost face displays an odd number of spots. Express your answer as a common fraction reduced to lowest terms.